

Exercise set 4

Textbook: Exercises 3.1, 3.11, 4.1.

Problem 1:

- a)^R Implement an algorithm to simulate arrival times for a homogeneous Poisson process with intensity λ over an interval $[0, t]$ where you make use of the following two properties of homogeneous Poisson processes:
- i) The number of arrivals during an interval of length t is Poisson distributed with expectation λt ,
 - ii) Given the number of arrivals n during the interval $[0, t]$ the arrival times are uniformly distributed over the interval.
- b)^R For a homogeneous Poisson process with intensity $\lambda = 2$, calculate $P(N(10) > 25)$ by simulation.
- c) Find the probability in b) analytically.